

May 14th 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
 - three syntheses done with 0.02M sulfuric acid
 - first product not representative, not used further
 - second and third batch converted (see below)
 - four syntheses done with 0.10M sulfuric acid
 - very similar observations and outputs
 - products of these have been combined/homogenized
 - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
 - first two synthesis batches finished, two more in progress (one in drying, one in hydrothermal stage)
- (Re)-Activation of larger vessel initiated

2. Biochar analysis

- First samples (TB1133, Sigma-Aldrich Reference) sent out for Raman / SEM

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
 - biochar ground manually for first experiment
 - laboratory mill used for second experiment
 - products not yet characterized
 - no serious handling issues
- third experiment performed with much gentler heating ramp (3 °C/min)
 - better handling but no other apparent difference

4. Graphene analysis

- details under discussion

Experiment overview

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2928	AV1	6.0 g BAFA neu Hemp Fibre VF	100.0 g 0.19% 0.02 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.4 bar	742 g Water	2.6 g	->										
Comment	Fibres not completely covered by liquid																		
MB2933	AV1	6.1 g BAFA neu Hemp Fibre VF	174.0 g 0.19% 0.02 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.5 bar	647 g Water	1.2 g	->	TB1131	1.2 g 1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.16 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MB2935	AV1	6.0 g BAFA neu Hemp Fibre VF	172.9 g 0.19% 0.02 M Sulfuric acid	180 °C	23 h	8.8 bar -> 9.5 bar	500 g Water	1.3 g	->	TB1132	1.3 g 1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MB2936	AV1	6.0 g BAFA neu Hemp Fibre VF	172.4 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.8 bar -> 10.6 bar	589 g Water	0.8 g	->	TB1133	1.0 g 1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MRa225	AV1	6.0 g BAFA neu Hemp Fibre VF	153.8 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	? -> 10.5 bar	500 g Water	0.9 g	->										
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MRa228	AV1	6.3 g BAFA neu Hemp Fibre VF	154.0 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 11.0 bar	500 g Water	0.9 g	->										
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MRa231	AV1	6.1 g BAFA neu Hemp Fibre VF	168.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.0 bar -> 10.5 bar	500 g Water	0.8 g	->										
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																		
MB2937	AV1	9.4 g Canadian Rockies Hemp	179.3 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 10.9 bar	569 g Water	1.6 g	->										
Comment	very low amount of foreign material observed and removed during filtration																		
MB2938	AV1	8.9 g Canadian Rockies Hemp	175.0 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.2 bar -> 13.2 bar	577 g Water	1.4 g	->										
Comment	very low amount of foreign material observed and removed during filtration																		
MB2940	AV1	9.2 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.6 bar	555 g Water		->										
Comment	very low amount of foreign material observed and removed during filtration																		
MB2942	AV1	8.9 g Canadian Rockies Hemp	177.6 g 0.96% 0.10 M Sulfuric acid	180 °C		->	Water		->										
Comment																			